(e) Current Aggregate Exchange Revenues from Selling Market Data and Connectivity

The Commission estimates that in 2018 the exchanges earned a total revenue of approximately $941 million from selling both proprietary and SIP market data products and connectivity services in the equities market. In addition, the Commission estimates that the exchanges earned approximately $596 million of this $941 million revenue from selling market data products and approximately $345 million of this revenue from selling connectivity services. With respect to the revenue from market data products, the Commission estimates that in 2018 the exchanges earned approximately $327 million of the $596 million revenue from equity SIP data and approximately $269 million from selling proprietary data products. Further, approximately $63 million of the $327 million equity SIP revenue in 2018 was distributed to FINRA.[1747]

The Commission's estimates above are mainly based on revenue information that the exchanges filed as part of their Form 1 filings.[1748] In addition, the Commission used SIP revenue information disclosed by the CTA/CQ Plans and the Nasdaq UTP Plan in their quarterly revenue disclosures.[1749] Furthermore, because revenue information provided by some exchanges in their Form 1 filings is not sufficiently detailed for this calculation, the Commission had to make

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1747 When taking this $63 million into account, total SIP revenues shared by SROs were approximately $390 million in 2018, which is consistent with the $430 million estimate for 2017 provided in the Proposed Governance Order (which also included the amount paid to the plan processor). See supra note 1728 and accompanying text. This estimate is also consistent with the $387 million estimate for 2017. See Jones Letter, supra note 1729 at 25.

1748 See Commission, National Securities Exchange Periodic Amendments to Form 1 (Modified June 20, 2019), available at https://www.sec.gov/rules/national-securities-exchanges-amendments.htm (providing links to exchanges' Form 1 filings).

1749 See supra note 1729.

certain assumptions in order to derive these estimates. First, the Form 1 filings for NYSE and NYSE MKT combine revenue from connectivity fees with revenue from market data fees. For these exchanges, the Commission derived the revenue earned from connectivity fees by assuming that the revenue that these exchanges earn from proprietary data is slightly smaller than the revenue that they earn from SIP data (based on notes in their Form 1 filings that indicate that SIP revenue exceeds proprietary data revenue). Second, the Form 1 filing for Nasdaq combines revenue from connectivity fees with revenue from transaction fees. The Commission derived the revenue that Nasdaq earned from connectivity fees by assuming that Nasdaq's revenues from connectivity fees and transaction fees were in the same proportion to one another as NYSE's revenues from these two business lines. Third, Form 1 filings for exchanges that offer trading in both equities and options provide revenue information for these two asset classes combined. For these exchanges, the Commission assumed that their combined revenues from market data fees and connectivity fees in the equities market and in the options market were in the same proportion to one another as the market data and connectivity revenues that these exchanges would have earned in each of these markets based on their dollar volume market share (as compared to the dollar volume market share of the exchanges that trade only equities or only options).

(f) Current State of National Best Bid or Offer Dissemination

Some commenters characterized the current process for dissemination of the NBBO as being based on a universally trusted source in the form of the exclusive SIPs, upon which all

market participants heavy rely.[1750] Commenters also suggested that the introduction of "multiple NBBOs" into the market would be a significant departure from current market practice.[1751] The Commission disagrees with this characterization of the relevant baseline for the final amendments. As mentioned in the Proposing Release,[1752] today, at any given instant of time, there can be differences between various market participants in what they observe to be the prevailing NBBO. These differences arise because of the geographic dispersion of the exchange data centers and the differences in latency between consolidated market feeds produced by the SIPs and those produced through the use of proprietary data feeds. Furthermore, the amount of time that typically elapses before the differences are corrected is meaningful to market participants.

Geographic latency means that even if all market participants relied on the exclusive SIPs, there would still be differences in what market participants observed to be the prevailing quote. For example, suppose the CTA/CQ SIP receives an update about the prevailing NBBO in a given stock. That information must still be disseminated to the various broker-dealers at the different data centers. At a minimum, there will be broker-dealers located in Mahwah, Secaucus and Carteret who will all be interested in seeing the new quote. The exclusive SIP distributes the quote to each of the data centers at the same time, but since these are in different locations, the quotes will arrive at different times.[1753] Therefore, as many as three different quotes for the same

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1750 See, e.g., Nasdaq Letter IV at 3.

1751 See, e.g., Nasdaq Letter IV at 11; TD Ameritrade Letter at 12.

1752 See Proposing Release, 85 FR at 16845.

1753 While they will likely all arrive within roughly a millisecond of each other, this is still a meaningful discrepancy in today's markets.

stock could be observed to be the NBBO in that stock at these three locations at a given instant in time, at least for market participants who are latency-sensitive enough to detect such differences.

On top of this basic geographic latency source of differing NBBOs, the latency differential that exists between NBBOs obtained from the exclusive SIP and NBBOs produced by consolidating proprietary feeds[1754] further contributes to the discrepancies in market views possessed by market participants.

Market participants have adapted to this state of affairs. For example, some of the concern in the market about obtaining fast market data is directly connected to the existence of multiple NBBOs.[1755] Market participants often use co-location in order to be closer to the trading center and thereby receive updates with less delay than they would experience if they were located elsewhere, in order to prevent themselves from acting on stale NBBO quotes that may be different from the NBBO prevailing at the trading center.[1756] In addition, the inspection and enforcement conducted by SROs with regard to best execution obligations has evolved to consider this phenomenon. Specifically, SRO inspections typically request data from a broker-

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1754 See supra Section V.B.2(b) for a discussion of the types of latencies.

1755 See supra Section V.B.2(b) for discussion of the value of speed in trading and data access. In that section, the Commission discussed the value of being faster than one's competitors. One way in which this is relevant is that if a competitor's order executes against the NBBO before some other competitor, then the second competitor's order will arrive at the trading center based on information about the NBBO that is no longer true at the time that the order arrives.

1756 See MEMX Letter at 6 stating that because inherent geographic differences "…market participants may each have a different view of market data and events based on where they are located and the technologies and telecommunication techniques used."

dealer in evaluating whether a violation has occurred.[1757] Also, the Commission has stated that for the Order Protection Rule a trading center "…will be assessed based on the times that orders and quotations are received, and trades are executed, at that trading center."[1758] This statement reflects the fact that the inevitable latency differential between two trading centers means that there may be multiple NBBOs in the market depending on which trading center one is at. Also, the lookback provision of Rule 611[1759] recognizes that an observed NBBO may not be the current prevailing NBBO.[1760] As detailed above, the potential for multiple NBBOs has been understood and dealt with for some time, and therefore should not be problematic for market participants.

3. Competition Baseline

This section discusses, as it relates to this rulemaking, the current state of the market for core and SIP data products, the market for proprietary data products, the market for connectivity services, and the market for trading services as well as broker-dealers' competitive strategies for trading services.

1757 See supra Section III.B.10(d). Also, FINRA Rule 4554 requires that ATSs report the NBBO in effect at the time of order execution and the timestamp of when the ATS captured the effective NBBO.

1758 See Regulation NMS Adopting Release, 70 FR at 37523, note 215.

1759 17 CFR 242.611(b)(8) (Rule 611(b)(8)) provides a one-second "window" prior to a transaction, which allows a trading center to trade at any price equal to or better than the least aggressive best bid or best offer displayed at another trading center during the previous second.

1760 See Regulation NMS Adopting Release, 70 FR at 37523.

(a) Current Structure of Market for Core and SIP Data Products

Under the Equity Data Plans, SIP data is collected, consolidated, processed, and disseminated by the exclusive SIPs.[1761] Equity Data Plan Operating Committees, which are composed of the SROs, set the fees the exclusive SIPs charge for SIP data.[1762] Any revenue earned by the exclusive SIPs, after deducting their operating costs and FINRA's OTC oversight costs, is split among the SROs. FINRA rebates a portion of the exclusive SIP revenue it receives back to broker-dealers based on the trade volume it reports.[1763] The nature of the Equity Data Plan Operating Committee's responsibilities can create a conflict of interest for the SROs, as discussed above.[1764]

Each Equity Data Plan selects a single exclusive SIP through a bidding process to be the exclusive distributor of the plan's data.[1765] This grants the SIP a monopoly franchise in the distribution of the plan's data, which means that the SIPs are not subject to competitive forces that would produce more efficient outcomes. The Commission acknowledges that some economic theory would point to the opposite conclusion, but does not believe that it applies here. In particular, a paper by Demsetz would predict that the current monopolistic structure is most efficient.[1766] In industries where there are economies of scale, a monopoly franchise structure

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1761 See Proposing Release, 85 FR at Section II.A, for discussion of these issues in the Proposing Release.

1762 See supra note 1726 and accompanying text.

1763 See supra note 1727.

1764 See supra Section V.A.2.

1765 See supra Section IV.A.

1766 See Harold Demsetz, *Why Regulate Utilities*?, 11 J. L. & ECON. 55 (1968) ("Demsetz (1968)").

may lead to the most efficient means of production. This profile applies to the distribution of core data because of the high fixed costs.[1767] Demsetz (1968) argues that just because an industry has a monopolistic provider of a service does not mean that it is not subject to competitive forces. In particular, Demsetz (1968) argues that if the monopolistic provider of a service is subject to competition in the bidding process it could provide sufficient competitive incentives to achieve a competitive outcome. However, many theories provide examples of situations in which the monopoly franchise structure is less efficient than other structures.[1768] The Commission does not believe that the exclusive SIP bidding process provides sufficient competitive incentives for two reasons. First, the bidding process could be subject to conflicts of interest since some of the SROs voting to select the exclusive SIP are also bidding to be the SIP. Second, the contracts are not bid out regularly, so there may not be a significant chance that the current exclusive SIP will be replaced. Therefore, the Commission does not believe that the bidding process for exclusive SIPs is likely to produce the most efficient outcome and subject the exclusive SIPs to competitive forces.

In the Proposing Release, the Commission stated that historically there were not a large number of bidders for SIP tenders, and listed this as one of three reasons why the Commission does not believe that the exclusive SIP bidding process provides sufficient competitive incentives in the above discussion. Since then, the Commission has come to understand that there were 11

1767 See infra note 1797 and accompanying text.

1768 See, e.g., Oliver E. Williamson, *Franchise Bidding for Natural Monopolies – in General and with Respect to CATV*, 7 BELL J. ECON. 73 (1976) (discussing why bidding for monopolies may not work well); Robin A. Prager, *Firm behavior in franchise monopoly markets*, 21 RAND J. ECON. 211 (1990).

bidders for the UTP tender offer in 2014.[1769] Based on this new understanding, the Commission no longer believes that the process bidding for SIP tenders may be hindered by the number of bidders. However, the first two reasons discussed, namely, conflicts of interest and lack of regular new bidding on the contract, are sufficient reasons for the Commission to continue to believe that the bidding process may not be adequately competitive. Thus, the Commission continues to believe that the conclusions of Demsetz (1968) do not apply in this case, as discussed above.

The exclusive SIPs have significant market power in the market for core and aggregated market data products and are monopolistic providers of certain information, which means that for all such products they would have the market power to charge supracompetitive prices. Fees for core data are paid by a wide range of market participants, including investors, broker-dealers, data vendors, and others.

One reason the exclusive SIPs have significant market power is that, although some market data products are comparable to SIP data and could be used by some core data subscribers as substitutes for SIP data in certain situations, these products are not perfect substitutes and are not viable substitutes across all use cases. For example, as mentioned above, some market data aggregators buy direct depth of book feeds from the exchanges and aggregate

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[1769] See UTP *Operating Committee Selects Nasdaq as Processor Announcement*, JORDAN & JORDAN, available at, https://www.jandj.com/sites/default/files/library/UTP_SIP_Processor_Announced_2014.pdf

them to produce products similar to SIP data.[1770] However, these products do not provide market information that is critical to some subscribers and only available through the exclusive SIPs, such as LULD plan price bands and administrative messages.[1771] Additionally, some SROs offer TOB data feeds, which may be considered by some to be viable substitutes for SIP data for certain applications.[1772] However, broker-dealers typically rely on the SIP data to fulfill their obligations under Rule 603 of Regulation NMS, i.e., the "Vendor Display Rule," which requires a broker-dealer to show a consolidated display of market data in a context in which a trading or order routing decision can be implemented.[1773]

The purchase of SIP data or proprietary market data from all exchanges, either directly or indirectly, is necessary for all market participants executing orders in NMS stocks.[1774] SROs

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1770 The feeds produced by market data aggregators offer additional features, such as lower latency, but usually cost more than SIP data. See Roundtable Day One Transcript at 126–29 (Mark Skalabrin, Redline Trading Solutions).

1771 See Proposing Release, 85 FR at Sections III.D, III.E. One commenter agreed, stating that because the exclusive SIPs are the sole source of such messages, many market participants must purchase both proprietary and exclusive SIP feeds. See MEMX Letter at 3.

1772 In the equity markets, the top of book feeds offered by the SROs are usually cheaper than SIP data. However, they may only contain information from one exchange, or one exchange family. See, e.g., Nasdaq Basic, supra note 1651; CBOE One, supra note 9 at n.19; NYSE BQT, https://www.nyse.com/market-data/real-time/nyse-bqt; TD Ameritrade Letter 2018, supra note 1651 (stating that the lower cost of exchange TOB products, coupled with costs associated with the process to differentiate between retail professionals and non-professionals imposed by the SIP Plans, and associated audit risk, favors retail broker-dealer use of exchange TOB products).

1773 See Vendor Display Rule, Rule 603 of Regulation NMS; Proposing Release, 85 FR at Section IV.B.2(a).

1774 For example, 17 CFR 242.611(a) (Rule 611(a)) of Regulation NMS requires trading centers to establish policies and procedures to prevent trade-throughs. In order to prevent trade-throughs, executing broker-dealers need to be able to view the protected quotes on

have significant influence over the prices of most market data products. For example, the exchanges individually set the pricing of the TOB data feeds that they sell to market data aggregators and broker-dealers that self-aggregate who in turn generate consolidated data. At the same time, SROs collectively, as participants in the national market system plans, decide what fees to set for SIP data.[1775] Although market data aggregators might compete with the exclusive SIPs by offering products that provide consolidated data, they ultimately derive their data from the exchanges' direct proprietary data feeds, whose prices are set by the exchanges, a subset of SROs.[1776]

Regarding the level of competition among non-SRO market data aggregators that sell consolidated data to market participants, the Commission currently does not have an estimate of the number of players in this market and does not know how specialized these players are.[1777]

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all exchanges. They can fulfill this requirement by using SIP data, proprietary data feeds offered by the SROs, or a combination of both.

1775 See supra Section V.B.1.

1776 Pursuant to Section 19(b) of the Exchange Act and Rule 19b-4 thereunder, SROs must file with the Commission proposed rules, in which they set prices for their direct feed data. Those prices can vary depending on the type of end user.

1777 The Commission understands that certain entities from the list of market data vendors published on Nasdaq's website currently perform the market data aggregator function. See Proposing Release, 85 FR at n.516. This list does not provide a lower bound on the number of such entities because the list includes firms that the Commission believes are unlikely to perform high-speed data aggregation. The list is also not an upper bound because the Commission does not believe that all firms performing market data aggregation are listed. While the Commission does not have the number of data aggregators, the Commission has analyzed the effects on such parties qualitatively. The Commission does not have this information because data aggregators are not required to register with the Commission and/or identify themselves as data aggregators. Additionally, the Commission requested this information and did not receive any comments providing estimates of the number of data aggregators. While Commission does not have quantitative information, the Commission does have the insights discussed